Week 11 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

Question 1

Given below is the statement: *"An API is a set of rules and protocols that allows different software applications to communicate with each other, while a programming language is a formal language used to write instructions for a computer to perform specific tasks."*

(a) True ✓

(b) False

(c) Can't say

Solution

Both halves of the statement are correct definitions:

- **API (Application Programming Interface):** a contract that says what requests one program may send to another and what responses it will get back. Example: the Twitter (X) API lets your program ask for a user's tweets and receive them as JSON.
- **Programming language:** a formal language with fixed syntax and meaning (Python, Java, C) used to write the instructions a computer executes.

They are different things that work together: you write code in a programming language, and that code calls an API to talk to another application. For example, a Python script can call the Facebook Graph API.

Answer: True.

Question 2

Which of the following features is supposed to be a part of tweet content on Twitter? (multiple correct)

(a) Presence of pronouns ✓

(b) No. of followers

(c) No. of retweets

(d) Mention of self words (I, my, mine) ✓

Solution

In studies of tweet credibility and fake content, features are grouped by where they come from:

Feature group	Examples
Content (tweet-based)	Words used, presence of pronouns, self words (I, my, mine), length, URLs, hashtags, punctuation like '!' and '?', sentiment
User (source-based)	Number of followers, number of friends, account age, verified status
Propagation / network	Number of retweets, replies, how the tweet spreads

Pronouns and self words can be read directly from the tweet text, so they are content features. The number of followers describes the account (user feature), and the number of retweets describes how others reacted (propagation feature); neither is part of the text.

Answer: Presence of pronouns, and Mention of self words (I, my, mine).

Question 3

Who among the following are most likely to engage in link farming? (multiple correct)

(a) Highly active users ✓

(b) Legitimate users ✓

(c) Popular users ✓

(d) Bloggers and experts ✓

Solution

A surprising finding of the link farming study on Twitter (Ghosh et al.) is that spammers get most of their follow-backs from genuine users, not from other spammers. These users are called social capitalists: they want to grow their own follower count and influence, so they follow back almost anyone who follows them.

The study found these users are typically:

- Highly active: they tweet a lot and interact often.
- Legitimate: they are real people, not spam accounts.
- Popular: they already have many followers (high indegree and high outdegree, as in Week 7 Question 10).
- Bloggers, experts and marketers who gain from a larger audience.

Because their follow-back habit lends reach to spammers, these users unknowingly help link farming.

Answer: All four: highly active users, legitimate users, popular users, bloggers and experts.

Question 4

Which of the following is username creation behaviour? (multiple correct)

(a) Static behavior patterns ✓

(b) Occasional reuse patterns

(c) Temporal behavior patterns ✓

(d) Frequent reuse patterns

Solution

The username change study on Twitter (covered in Week 10) splits user behaviour into two broad groups:

Group	Pattern	Meaning
Creation (how new names are made)	Static behaviour	The new username follows a fixed style or keeps a common core (for example the same base name with small edits)
Creation (how new names are made)	Temporal behaviour	The username changes with time or events (for example adding a year, season, festival or trending event)
Reuse (going back to old names)	Occasional reuse	A user sometimes returns to a username used before
Reuse (going back to old names)	Frequent reuse	A user keeps cycling back to old usernames again and again

Questions 4 and 5 ask for the two halves of this table. Static and temporal patterns describe how new usernames are created.

Answer: Static behavior patterns, and Temporal behavior patterns.

Question 5

Which of the following is a username reuse behaviour? (multiple correct)

(a) Static behavior patterns

(b) Occasional reuse patterns ✓

(c) Temporal behavior patterns

(d) Frequent reuse patterns ✓

Solution

From the table in Question 4, reuse behaviour means a user returns to a username they held earlier. Depending on how often this happens, it is called occasional reuse or frequent reuse.

Reuse matters for security: while a user's old name is free, someone else can grab it (username squatting) and impersonate them. The old name also no longer leads to the user's profile.

Answer: Occasional reuse patterns, and Frequent reuse patterns.

Question 6

Why is it called a "Graph" API?

(a) It allows users to draw and create graphs and charts.

(b) It represents social connections and relationships as a graph data structure. ✓

(c) It primarily deals with geographical mapping and location-based services.

(d) It is used exclusively for graph theory and mathematical computations.

Solution

Facebook's Graph API models all of Facebook as a social graph, made of three building blocks:

- **Nodes:** objects such as users, pages, photos, posts and comments, each with an ID.
- **Edges:** connections between nodes, such as a user's friends, a page's posts or a photo's comments.
- **Fields:** properties of a node, such as a user's name or a post's time.

A request like GET /{user-id}/friends follows the 'friends' edge from a user node. The word 'graph' refers to this nodes-and-edges data structure, not to drawing charts, maps or doing mathematics.

Answer: It represents social connections and relationships as a graph data structure.

Question 7

Latanya Sweeney's research highlighted how two seemingly unrelated pieces of information can be linked to reidentify an individual. Which term best describes this concept?

(a) Data Anonymization

(b) Data Aggregation

(c) Reidentification ✓

(d) All of the above

Solution

In the late 1990s, Massachusetts released 'anonymised' hospital records of state employees, with names removed but ZIP code, date of birth and sex kept. Sweeney bought the public voter list of Cambridge, which contains names together with the same three fields. By linking (joining) the two datasets on ZIP + date of birth + sex, she found the medical record of the state Governor, William Weld.

She also showed that ZIP code, date of birth and sex together uniquely identify about 87% of the US population. Such fields are called quasi-identifiers.

Recovering a person's identity from data that was supposed to be anonymous is called reidentification (a linkage attack). Anonymisation is what was attempted and failed, and aggregation (combining data into summaries) is a different process, so (a), (b) and (d) are not the best answer.

Answer: Reidentification.

Question 8

What is the primary purpose of an API key on Twitter?

(a) To limit the number of characters in a tweet.

(b) To verify a user's identity on the platform.

(c) To authenticate and access Twitter's APIs for development purposes. ✓

(d) To encrypt and secure direct messages on Twitter.

Solution

When a developer registers an app on the Twitter (X) developer portal, they receive an API key and secret (and access tokens). Every request the program sends includes these credentials so that Twitter can:

- Identify which application is calling (authentication).
- Check what the app is allowed to do (authorisation).
- Apply rate limits and track usage for that app.

The key identifies the developer's application, not an ordinary user logging in (b). It has nothing to do with tweet length (a) or encrypting messages (d).

Answer: To authenticate and access Twitter's APIs for development purposes.

Question 9

What is the primary purpose of link farming in the context of SEO (Search Engine Optimization)?

(a) To create high-quality, relevant backlinks to improve a website's search engine ranking.

(b) To increase organic traffic by optimising on-page content and meta tags.

(c) To manipulate search engine algorithms by generating a large number of low-quality backlinks. ✓

(d) To enhance the user experience by improving website navigation and design.

Solution

Search engines such as Google (PageRank) treat a link from another site as a vote of confidence: more incoming links (backlinks) usually means a higher rank. A link farm is a network of websites that exist mainly to link to each other or to a target site, creating a large number of artificial, low-quality backlinks. The aim is to trick the ranking algorithm, not to provide real value, so it is a black-hat SEO technique. Search engines penalise sites caught doing this.

Option (a) describes genuine link building (white-hat SEO), (b) is on-page SEO, and (d) is user-experience design. None of them is link farming.

Answer: To manipulate search engine algorithms by generating a large number of low-quality backlinks.

Question 10

Which V of social media represents "Viral trends that spread rapidly across social media platforms"?

(a) Value

(b) Velocity ✓

(c) Volume

(d) Veracity

Solution

Social media data is a form of big data, often described by its V's:

V	Meaning in social media
Volume	The huge amount of data (billions of posts every day)
Velocity	The speed at which data is created and spreads, e.g. a hashtag going viral within minutes
Variety	Different forms of data: text, images, video, location
Veracity	Trustworthiness of the data: fake news, rumours, spam
Value	Useful insight that can be extracted from the data

The key word in the question is 'rapidly'. Speed of spread is velocity. For example, the Boston blast tweets in Week 10 reached about 2 lakh tweets per hour within the first hour.

Answer: Velocity.

Answer Key

Q.No.	Correct answer
1	True
2	Presence of pronouns; Mention of self words
3	All four options
4	Static behavior patterns; Temporal behavior patterns
5	Occasional reuse patterns; Frequent reuse patterns
6	Represents social connections as a graph data structure
7	Reidentification
8	To authenticate and access Twitter's APIs for development
9	To manipulate search engines with many low-quality backlinks
10	Velocity